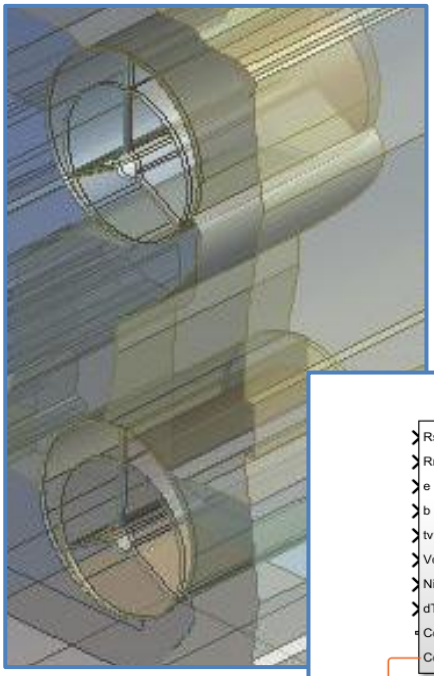
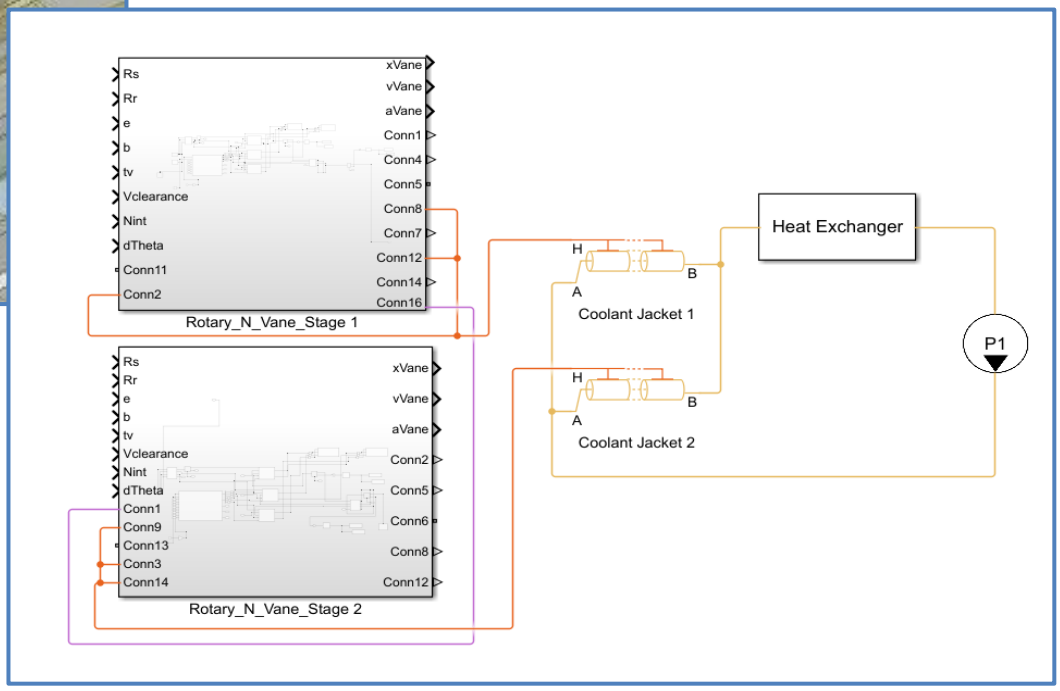
ROTARY VANE VACUUM PUMP

Cooling System Optimization

Oil-Sealed Two-Stage Rotary Vane Vacuum Pump

DOCUMENT PURPOSE: Technical record of the engineering work, architecture, investigations, and open technical items. It is written as a development report.





1. Executive Summary

This document summarizes the comprehensive thermal-fluid simulation, geometry & system optimization, and parameter modeling for a two-stage rotary vane vacuum pump operating within an advanced industrial cooling loop. The primary design objective is to achieve active thermal management across both compression stages, reducing localized thermal distortion, and optimizing overall performance.

Component	Details
Pump Architecture	Two-Stage Rotary Vane Vacuum Pump
Cooling System	Dual Integrated Water Jackets
Lubrication System	Fresh-Oil Lubrication Layout with Injection Network

The recent development work focused on two connected engineering tracks for an oil-sealed, two-stage rotary vane vacuum pump: (1) establishing a physically meaningful dynamic model of the rotor–vane–chamber system, and (2) investigating the pump cooling architecture and the appropriate scope for a cooling-system design study.

The cooling-system investigation considered the distinction between oil lubrication/sealing and external heat rejection, including water jackets, through water cooling, closed-loop radiator-fan circuits. The actual architecture of the selected pump still requires confirmation from its manufacturer documentation or a verified parts/circuit drawing.

KEY STATUS: A technical modeling architecture and investigation path were developed.

2. Two-Stage Pump System Representation

The portfolio scope identifies an oil-sealed two-stage rotary vane vacuum pump. A system-level model should therefore distinguish the low-vacuum (first) stage from the high-vacuum (second) stage and represent the inter-stage connection explicitly.

- Stage 1: suction-side chamber filling, compression, and transfer toward the inter-stage region.
- Inter-stage region: connecting volume and any relevant pressure losses or cooling/condensation effects supported by the actual pump design.
- Stage 2: further compression and discharge to the exhaust side.
- The exact stage arrangement, ports, internal bypasses, oil injection locations, and inter-stage cooling provisions must be taken from the selected pump's drawings.

3. Cooling-System Architecture Investigation

The thermal-system discussion examined what actually removes heat from an oil-sealed rotary-vane vacuum pump and whether water cooling is necessarily present. The key distinction is between oil's internal roles and the external heat-rejection mechanism.

Architecture	Heat-rejection path	Evidence/status
Air-cooled housing / fins	Heat passes from housing surfaces to ambient air, by natural or forced convection.	Candidate architecture; confirm for selected pump.
Water jacket	Coolant flows through passages around the pump body or selected hot regions.	Discussed; pending as fitted to the selected machine.
Once-through water cooling	External water supply passes through a heat exchanger or jacket and is discharged.	Possible arrangement; requires manual/utility confirmation.
Closed-loop radiator and fan	Coolant circulates through the pump-side exchanger and an air-cooled radiator.	Candidate circuit; not confirmed.

3.1 Key engineering conclusion

An oil-sealed pump is not automatically a water-cooled pump. The presence of oil does not prove the presence of a water circuit, and a water-jacket or oil cooler should not be included in the portfolio design as an existing feature unless it is supported by the selected pump's manual, parts list, P&ID, or manufacturer drawing. Some machines reject heat primarily through the oil and casing, while others use an external cooling arrangement.

3.2 Information needed to finalize the cooling system

- Manufacturer and exact model designation of the two-stage pump.
- Manufacturer cooling requirements and allowable operating/ambient temperatures.
- Whether the pump has a water jacket, oil cooler, or air-cooled heat exchanger.
- Oil flow rate, oil temperatures, and available pressure drop (if an oil cooler exists).
- Coolant type, supply temperature, flow rate, pressure, and utility constraints.
- Heat load or a defensible estimate based on motor input, compression work, and measured temperature data.

4. Verification and Validation Plan

Work package	Verification method	Acceptance evidence	Current status
Geometry & kinematics	Compare vane-tip position and chamber geometry against CAD/drawing at multiple rotor angles.	Position/volume curves agree with geometric reference.	Analytical approach discussed; verification pending.
Two-stage system	Check stage pressures, inter-stage flow, and mass balance.	Physically consistent pressure progression and stable initialization.	System validation done.
CFD	Mesh independence, time-step sensitivity, mesh-quality monitoring, and comparison with published/experimental data.	Repeatable pressure/velocity fields and chamber-pressure history.	Workflow investigated; simulation runs confirmed.
Cooling circuit	Thermal balance, component sizing, pressure-drop calculation, and temperature measurement.	Required heat duty removed with acceptable temperatures and flow resistance.	System Optimized. (8% Improvement) / Validation test pending